

**Problem 1:** SJF (Non – Preemptive):

**Source Code:**

```
#pragma GCC optimize("Ofast")

#include <bits/stdc++.h>

using namespace std;

#ifdef ONLINE_JUDGE

#define debug(x) cerr << #x << " = " << x << endl

#else

#define debug(x)

#endif

using ll = long long;

void Solve()
{
    int n;

    cin >> n;

    cout << "Total Process: " << n << endl;

    deque<pair<int, int>> proc(n);

    map<int, int> mp;

    for(int i=0; i<n; i++) {

        // cout << "\nBurst time of process " << i+1 << ": ";

        cin >> proc[i].first;

        proc[i].second = i+1;

        int x ;

        cin >> x;

        mp[i+1] = x;

    }
```

```

vector<int> ans;

int tem = 0;

ans.push_back(proc[0].first);

int p = proc[0].second;

cout << "\nProcess " << proc[0].second << " : ";

cout << "\nWaiting Time: " << tem << endl;

// cout << "Arrival Time: " << mp[proc[0].second] << endl;

tem += ans.back();

int TT = tem - mp[p];

cout << "Turn Around Time: " << TT << endl;

// cout << "WT: " << tem - proc[0].first << endl;

proc.pop_front();

sort(proc.begin(), proc.end());

int piche = proc.back().first;

while(!proc.empty()){

    cout << "\nProcess " << proc.front().second << " : " << endl;

    cout << "Waiting Time: " << tem << endl;

    tem += proc.front().first;

    int tt = tem - mp[proc.front().second];

    cout << "Turn Around Time: " << tt << endl;

    // cout << "WT: " << tem - proc.front().first << endl;

    proc.pop_front();

}

int av = tem - piche;

cout << "\nAverage Waiting Time: " << (double)av/n << endl;

```

```

}

int32_t main()
{
    ios_base::sync_with_stdio(0), cin.tie(0), cout.tie(0);

    ll t = 1;

    // cin >> t;

    while (t--)

        Solve();

    return 0;
}

```

Sample Input:

```

Input:
4
7 0
4 1
1 2
4 3

```

Sample Output:

```

Total Process: 4

Process 1 :
Waiting Time: 0
Turn Around Time: 7

Process 3 :
Waiting Time: 7
Turn Around Time: 6

Process 2 :
Waiting Time: 8
Turn Around Time: 11

Process 4 :
Waiting Time: 12
Turn Around Time: 13

Average Waiting Time: 3

```

**Problem 2: SJF non-preemptive****Source Code:**

```
#pragma GCC optimize("Ofast")

#include <bits/stdc++.h>

using namespace std;

#ifdef ONLINE_JUDGE

#define debug(x) cerr << #x << " = " << x << endl

#else

#define debug(x)

#endif

using ll = long long;

void Solve()
{
    int n;

    cin >> n;

    deque<pair<int, int>> v(n);

    for(int i=0; i<n; i++){
        cin >> v[i].first;
        v[i].second = i;
    }

    priority_queue<pair<int, int>, vector<pair<int, int>>, greater<pair<int, int>>> pq;

    map<int, int> mp;

    map<int, int> arrival;

    map<int, int> waiting;
```

```
map<int, int> turnaround;
```

```
mp[0] = 0;
```

```
v.front().first--;
```

```
for(auto it : v) pq.push(it);
```

```
int time = 0;
```

```
for(int i=0; i<n; i++){
```

```
    arrival[i] = 0;
```

```
}
```

```
while(!pq.empty()){
```

```
    auto it = pq.top();
```

```
    pq.pop();
```

```
    if(it.first == 0) continue;
```

```
    time++;
```

```
    mp[it.second] = time;
```

```
    it.first--;
```

```
    if(it.first > 0){
```

```
        pq.push(it);
```

```
    }
```

```
}
```

```

cout << "Process ID | Burst Time | Completion Time | Waiting Time | Turnaround Time\n";
cout << "-----\n";

int totalWaiting = 0;
int totalTurnaround = 0;

for(int i=0; i<n; i++){
    int burstTime = v[i].first;
    int completionTime = mp[i];
    waiting[i] = completionTime - burstTime - arrival[i];
    turnaround[i] = completionTime - arrival[i];

    cout << "P" << i << "    | " << burstTime << "    | " << completionTime
        << "    | " << waiting[i] << "    | " << turnaround[i] << "\n";

    totalWaiting += waiting[i];
    totalTurnaround += turnaround[i];
}

cout << "-----\n";
cout << "Average Waiting Time: " << (double)totalWaiting/n << "\n";
cout << "Average Turnaround Time: " << (double)totalTurnaround/n << "\n";
}

int32_t main()
{

```

```

ios_base::sync_with_stdio(0), cin.tie(0), cout.tie(0);

ll t = 1;

while (t--)

    Solve();

return 0;
}

```

**Input:**

Input: Copy

```

4
7
4
1
4

```

**Output:**

Received Output: Set Copy

| Process ID | Burst Time | Completion Time | Waiting Time | Turnaround Time |
|------------|------------|-----------------|--------------|-----------------|
| P0         | 6          | 15              | 9            | 15              |
| P1         | 4          | 5               | 1            | 5               |
| P2         | 1          | 1               | 0            | 1               |
| P3         | 4          | 9               | 5            | 9               |

-----

Average Waiting Time: 3.75  
Average Turnaround Time: 7.5